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UNIT I
INTRODUCTION AND NATURAL RESOURCES

- Renewable and non-renewable resources. Natural resources and associated problems.
- Forest, Water, Mineral, Food, Energy, and Land resources.
- Role of an individual in the conservation of natural resources.
- Equitable use of resources for sustainable lifestyles.

NATURAL RESOURCES

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Natural resources: Forest, Water, Mineral, Food, Energy, Land resources

Type of Energy Resources:

1) Renewable: Regenerates as fast as they are consumed and are continuously available, Eg: Solar, wind, water, geothermal
2) Non-Renewable: cannot be replenished. Eg:, Fossil Fuel and Nuclear materials.

Renewable Energy Source	Installed Capacity (MW) as on 2017
Solar	16611
Wind	32746
Biomass	8181
Small Hydro Power	4399
Waste to Energy	114

FOREST RESOURCES

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Forest, a green blanket on earth, provides a range of products & ecological services due to the diversity inherent in them.

Commercial Use:

- Timber
- Fuel Wood
- Raw Material
- Building materials
- Pulp & paper industry
- Bio-mass
- Recreation-aesthetic & educational tourism

Ecological function

- Productive
- Protective: Soil erosion, droughts, Floods
- Regulative function:
- Intensive radiations
- Noise

OVER EXPLOITATION OF FOREST

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Causes

- Population growth
- Clearing forests for agriculture
- Timber extraction
- Mineral extraction
- Dam construction and hydropower projects
- Medicinal products extraction
- Overgrazing

Effects

- Soil erosion and flooding
- Ecological imbalance
- Change in rain and season pattern
- Desertification
- Deforestation

DEFORESTATION

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Major activities in the forest lead to deforestation

- Timber extraction, Mining, Dams construction

Researchers investigate mining-related deforestation in the Amazon – University of Wisconsin-Madison

- Surface mining
- Sub-surface mining
- Require removal of vegetation along with underlying soil mantle and overlying rock masses. This results in defacing the topography and the landscape in the area
- Coal mining in Jharia, Raniganjj, and Singrauli in Jharkhand.
- Mining of radioactive material in Kerala, T.N and Karnataka
- Western ghats for excavation of Cu, Cr, bauxite, and magnetite.
- 80,000 ha are under stress of mining in India
- People living downstream suffer from soil and water contamination
- Workers suffer from skin and lungs diseases

Dams

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- ‘TEMPLES OF MODERN INDIA”
- Multipurpose drinking, irrigation, and hydropower
- **Socio-economic problems**: 40 to 80 million people have been displaced by 45,000 dam projects in recent years.
- Chipko Movement
- Narmada bacho Andolan
- The Narmada project consists of 30 large, 135 medium, and 3,000 minor dams on the Narmada River and its tributaries.
- It uprooted one million people, submerge 350,000 hectares of forest, drown 200,000 hectares of cultivable lands, and cost $52.2 billion.
- The **Indira Sagar Dam,** Narmada project promises to submerge 91,348 hectares of land and displace people from 254 villages.

https://fore.yale.edu/World-Religions/Hinduism/Engaged-Projects/Chipko-Movement

EFFECT OF DEFORESTATION

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- Destruction of natural habitats: Loss of Biodiversity
- Genetic diversity is eroded
- Nutrient and Hydrological cycle (rainfall) is affected
- Lose of soil fertility, Soil erosion
- In hilly regions - Landslides
- Imbalance in the food chain and nutrient cycle

- **Compensatory afforestation'**

"Compensatory afforestation" is the term used to describe the trees that must be planted as compensation for forests cut down, which requires special permission from the Ministry of Environment and Forests.

- **Total tree cover loss in India (2001-2018) - 19.1%**

Sustainable forestry

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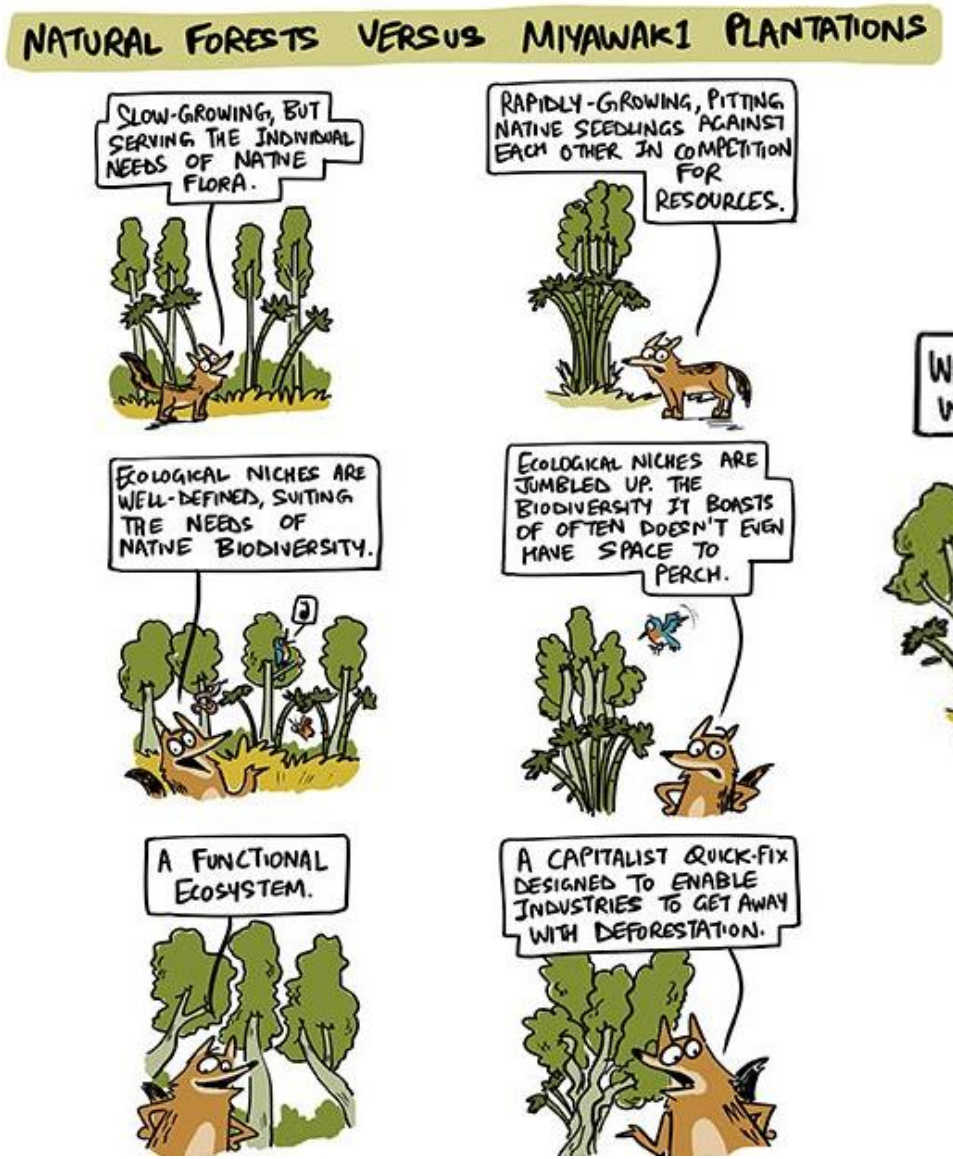
- "Sustainable forestry means managing our forests to meet the needs of the present without compromising the ability of future generations to meet their own needs by practicing a land stewardship ethic (cooperative planning and management of environmental resources).
- It integrates the reforestation, managing, growing, nurturing, and harvesting of trees for useful products with the conservation of soil, air, and water quality, wildlife and fish habitat, and aesthetics".



WATER RESOURCE

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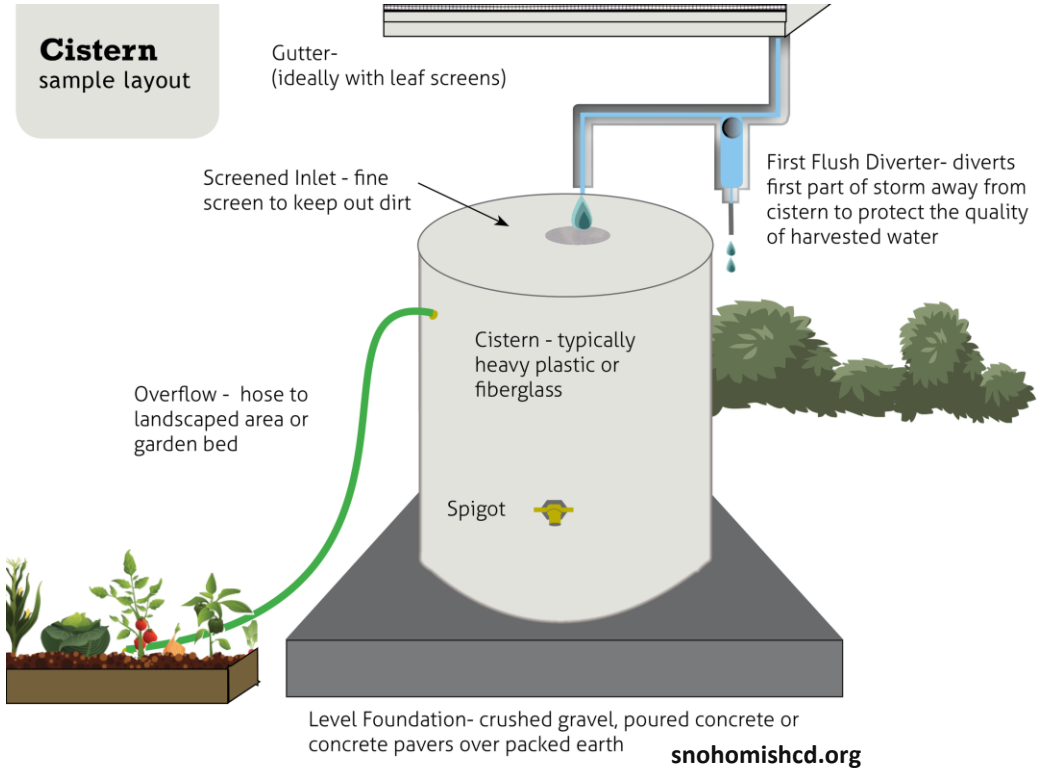
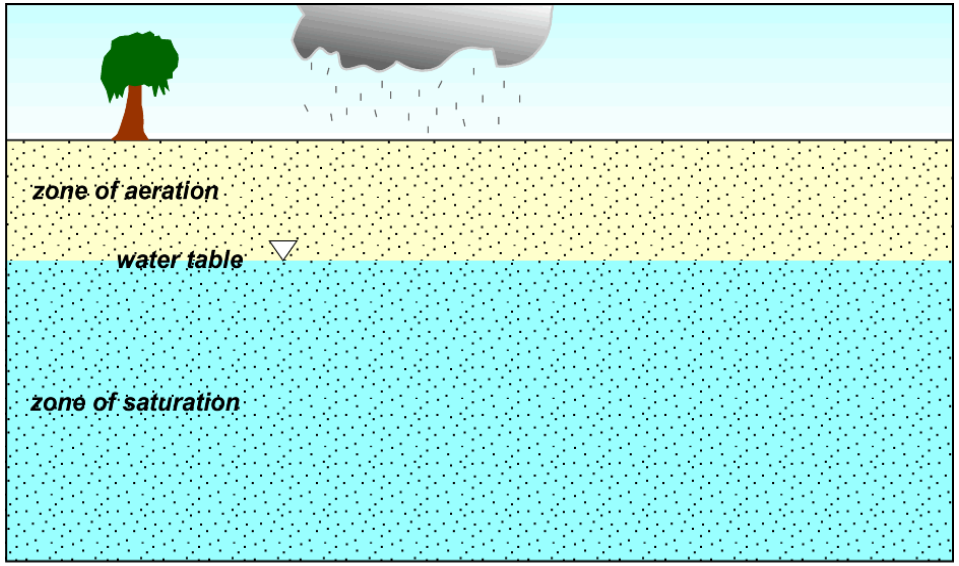
Sources of water

- Rain
- Surface water: Streams, Lakes, Ponds, Rivers, *Cisterns*
- Ground water: Open wells, Tube wells, Artesian wells, springs, Aquifers





Water table: the topmost surface of the water-saturated soil

AQUIFERS

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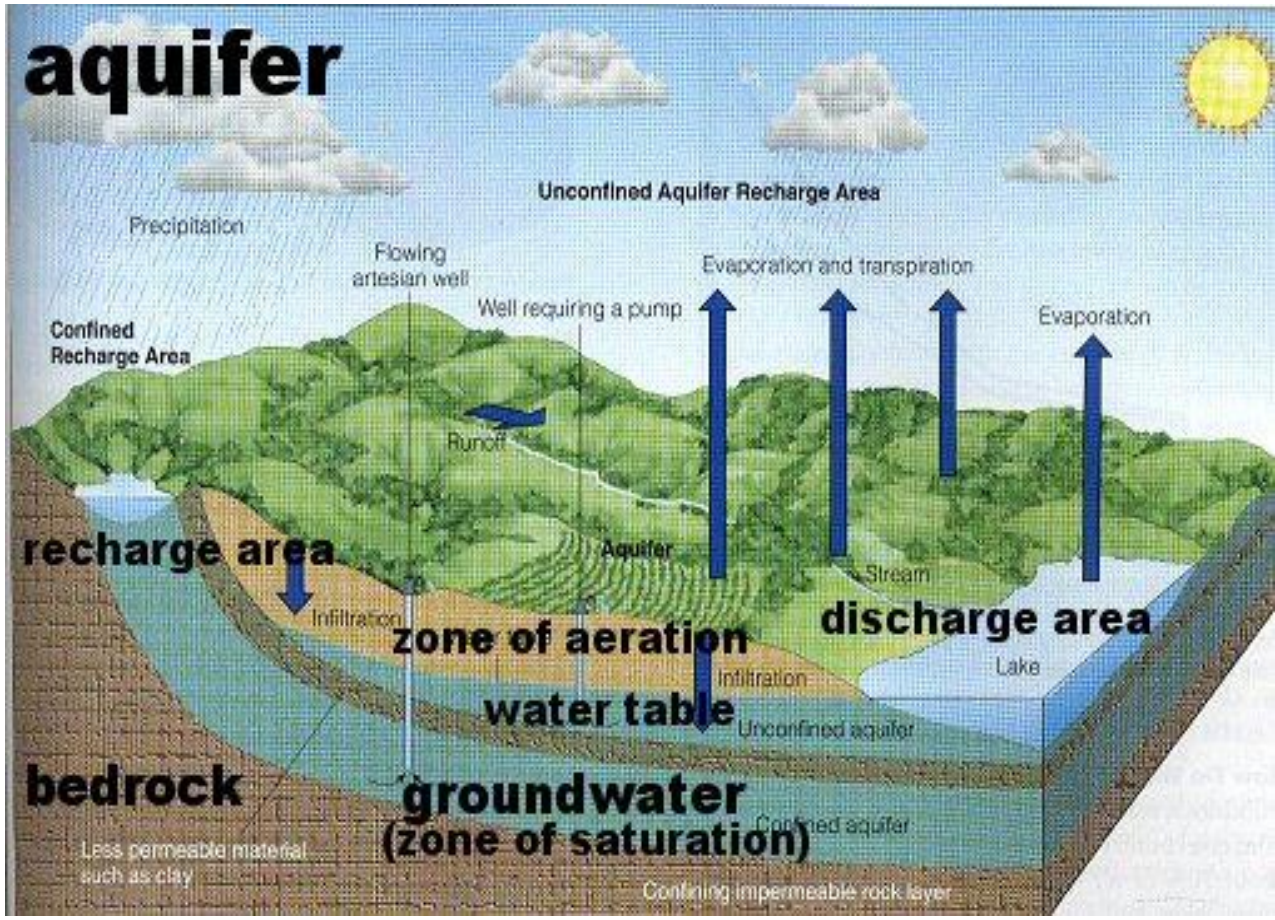
- Aquifer: A unit of permeable sediment underground layer that stores water and allows water to move through the pore spaces
- Infiltration is the entrance of rainwater into the soil; after Infiltration, water percolates in a downward direction till it reaches an impervious layer called *aquiclude*.

- Unconfined aquifer - overlaid by permeable rock or sediment & rechargeable

- Confined aquifer- sandwiched between two impermeable layer & rechargeable only where intersects with land surface



Book: Todd, D. K., & Mays, L. W. (2004). *Groundwater hydrology*. John Wiley & Sons.

WELLS

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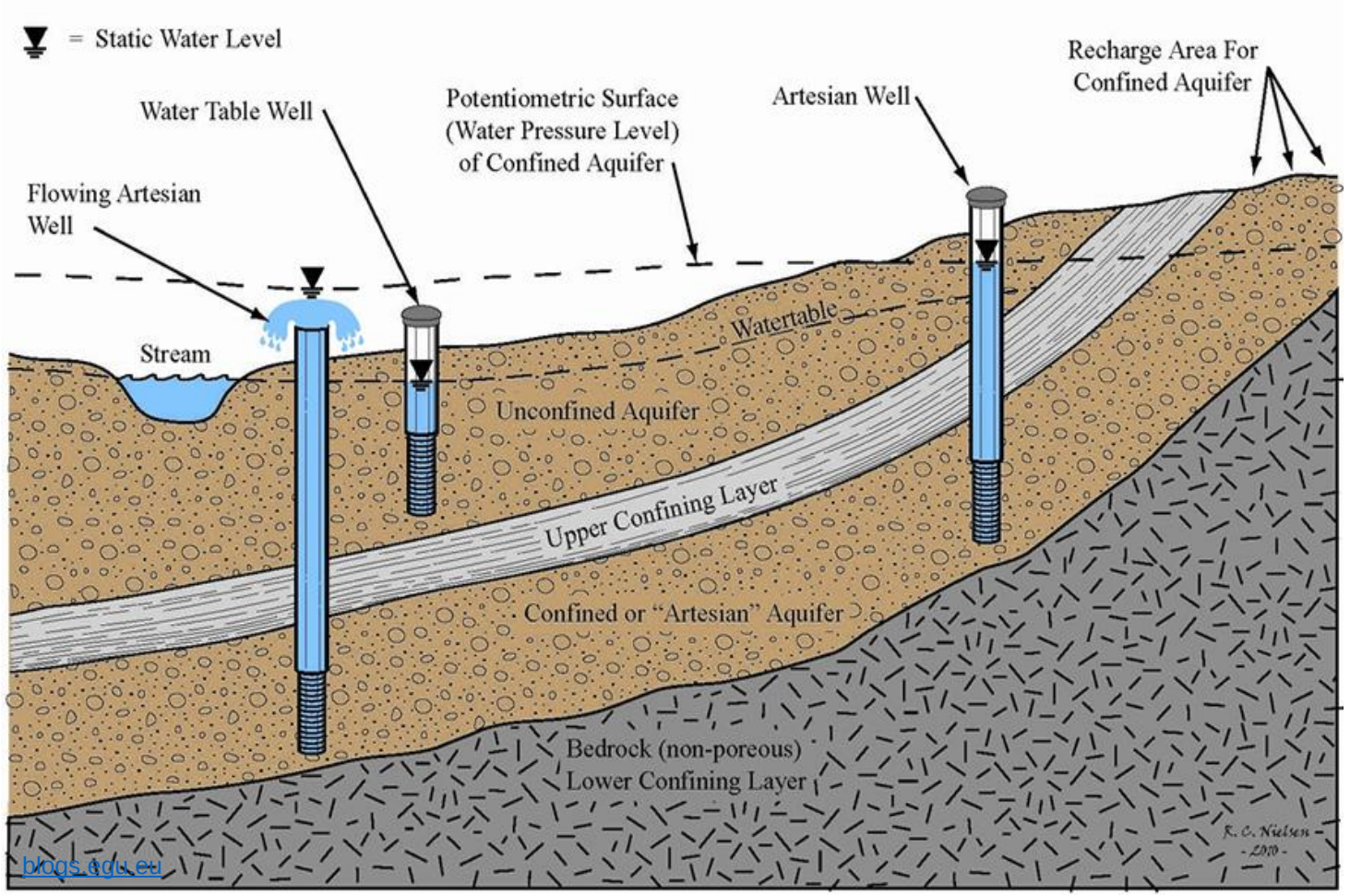
- An open well is an open vertical hole in the ground in order to bring water up using a pulley system
- A tube well is a long pipe that is sunk in the ground having one or more water-bearing strata
- An artesian well is a like a water-storing well that uses water pressure to bring water to the surface
- An infiltration well has filters that filter all the dangerous materials from water



Book: Todd, D. K., & Mays, L. W. (2004). *Groundwater hydrology*. John Wiley & Sons.

USES OF WATER

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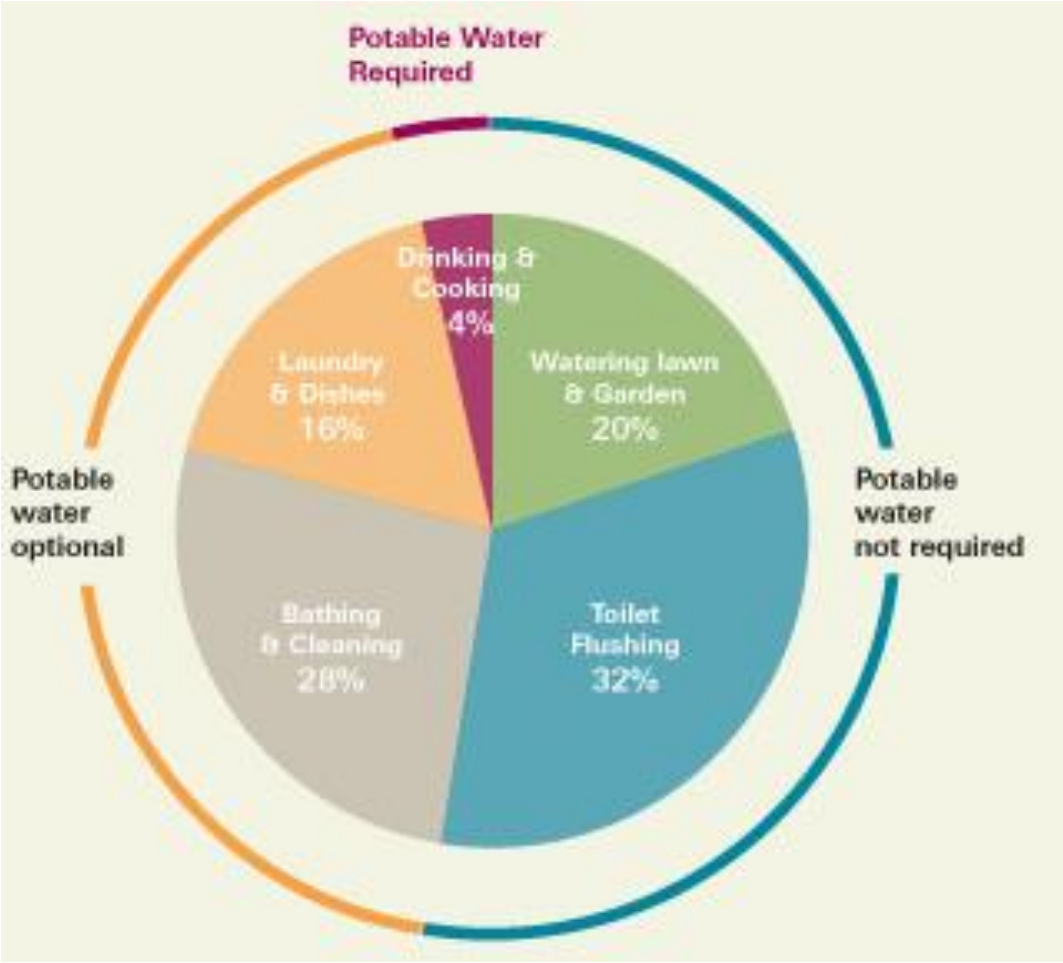
- **Municipal : Domestic purposes, Commercial**
- **Irrigation**
- **Fire demand**
- **Industrial purposes**

- **Non- potable**
- **Potable**



- **Problems associated with water resources**
- Over exploitation
  - By Water withdrawal
  - By Water consumption
- Contamination/ Water pollution
- **Floods and Drought**
- **Dams Construction Related Problems**
- **Conflicts Over Water**

EFFECT OF GROUNDWATER EXPOLTATION

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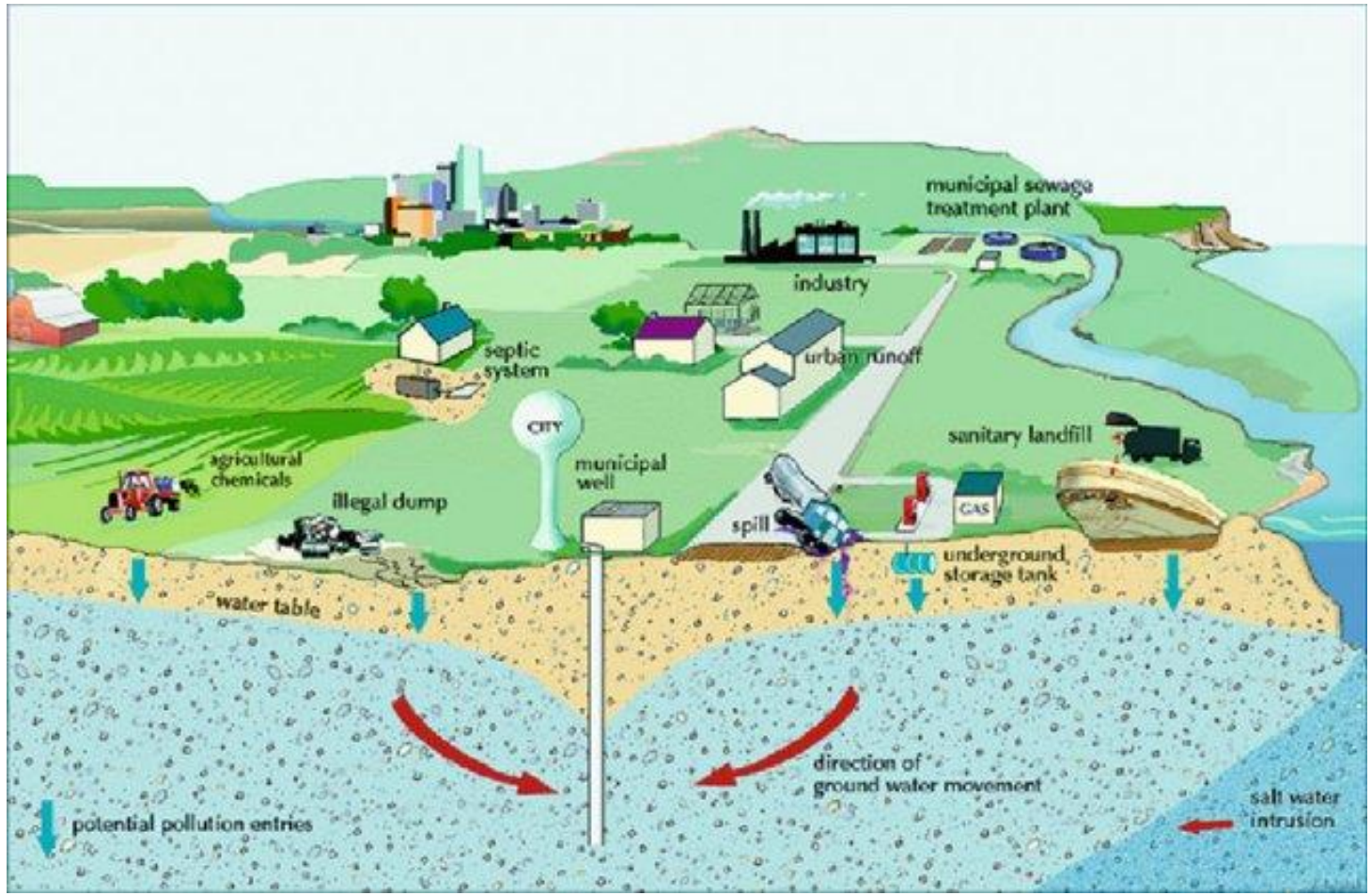
- Ground Subsidence
- Lowering the water table
- Water logging
- Groundwater contamination

Case 1 Joshimath: The trauma of living in India's sinking Himalayan town.

Case 2 : The worst change in land level occurred in North Jakarta with land subsidence estimated at around 4.9cm a year. The research found the main cause of the problem is overuse of groundwater by home drilling as a result of massive developments.



Book series: The Handbook of Environmental Chemistry Series, Volume title: The Nile Delta: chapter 6, Publisher: Springer

FLOODS

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- Overflow of water causing an expanse of water-submerged land

Main causes:

1. Heavy rainfall in low lying area and areas without proper drainage
2. Melting of ice
3. Coastal flood-Tsunami: a series of water waves (called a tsunami wave train) caused by the displacement of a large volume of a body of water, usually an ocean

- **Types of Floods**: Flash flood, River flood, Coastal flood
- **Consequences of floods:**

loss of lives and property, Soil erosion, Loss of soil fertility

- **Flood management:**

Dam construction, catchments, Protection by wetlands, Controlling runoff
Increasing the coping capacity of the people with emergency planning, warning
Compensation and insurance

- *Chennai flood, 2016*
- *Uttarakhand flood in 2013*

North Indian Flood in 2023
Flooding in the Netherlands and how are they tackling

DROUGHT

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- extended period where water availability falls below the statistical requirements for a region

Consequences of drought:

Desertification, Famine, Social and economic drop

Droughts can be of three kinds:-

(i) **Meteorological drought**: Actual rainfall in an area is significantly less than the climatological mean of that area.

(ii) **Hydrological drought:** Drying water reservoirs. A marked depletion of surface water causing very low stream flow and drying of lakes, rivers and reservoirs

(iii) **Agricultural drought:** Inadequate soil moisture resulting in acute crop stress and fall in agricultural productivity

Drought management:

- Water conservation, Dams, and lakes construction
- Using alternative technology for irrigation - drip irrigation
- Networking of rivers
- Social forestry method: the Kolar district of Karnataka is social forestry under world bank aid.
- Drought by Eucalyptus plantation- due to the high transpiration

DAMS –BENEFITS AND PROBLEMS

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- Ecological Benefits: : Prevention of drought, flood, improve the productivity
- Ecological Problems: Deforestation, loss of biodiversity, water logging, Flash floods, and reservoir and vector-born seismicity
- Socio-economic Benefits: employment, electricity generation, irrigation, navigation, fisheries, and drinking water
- Socio-economic Problems: submergence of villages & fertile lands, displacement, resettlement issues and the outbreak of vector born diseases, rehabilitation

Upstream Problems:

1. Displacement of native people
2. Loss of forest, flora and fauna
3. Siltation and sedimentation of reservoirs.
4. Water stagnation and water logging near the reservoir
5. Breeding of vector born diseases
6. Reservoir induced seismicity causing earthquakes.
7. Microclimate changes.

Downstream Problems:

1. Problem in getting water for need
2. Water logging and salinity due to over-irrigation
3. Breeding of vector-borne diseases
4. Change in oxygen level and temperature

CONFLICTS OVER WATER ISSUES

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- Unequal distribution of water often leads to interstate or international disputes.
- River crossing boundaries between countries or states can cause tension between the two.
- Over 200 water resources are shared between two countries and two states.
- In India, 17/18 major rivers are shared between two states.
- **Water stress** occur when the demand for water exceeds the available amount during a certain period or when poor quality restricts its use.

1. Middle east (Jordan, Syria, Israel and Turkey, Egypt, and Ethiopia over Nile and Tigris - Euphrates)
2. Indus water treaty: India (Satluj, Ravi, and Beas) and Pakistan (Indus, Jhelum & Chenab)
3. Brahmaputra River: India, China, and Bangladesh
3. Cauvery water dispute: Tamil Nadu & Karnataka
4. Satluj-Yamuna link dispute: Punjab & Haryana

WATER MANAGEMENT

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- Sustainable and wise usage of water resources
- Avoid water pollution
- Three R's – reduce, reuse, and recycle
- Using effective technology
- Rainwater Harvesting
- Judicious use
- Traditional water management system

LAND RESOURCES

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- **The land** is a finite and valuable resource upon which we depend for our basic amenities of life.
- Soil is a mixture of inorganic and organic material produced from rocks (parent material) through the processes of **weathering** and natural erosion.
- Water, wind, temperature change, gravity, chemical interaction, living organisms, and pressure differences all help break down parent material.
- **Topsoil** is a renewable resource and is regenerated by the natural process at a slow rate (formation of 1-inch soil takes 200-1000 years).
- Soil erosion is 20-100 times more than the rate of renewal
- **Land degradation** is defined as the decline in land quality or reduction in its productivity. The loss of potential productivity and utility as a result of natural or anthropogenic factors

LAND DEGRADATION

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Causes of Land degradation

1. Over-grazing
2. Deforestation
3. Soil Erosion: Geologic erosion, accelerated erosion
4. Mining
5. Water logging and salination,
6. Solid waste dumping,
7. Pollution

Process of Land degradation

1. Physical process decline of soil structure and soil texture by crushing, compacting, erosion, desertification, quarrying rocks, and sand quarrying
2. Chemical process – degradation by chemical processes such as acidification, leaching, alkalinity, envr. pollution and fertility depletion
3. Biological process – decrease in biomass and land biodiversity, contamination of water sources such as eutrophication.

SOIL EROSION

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Climatic agents

- Water
- Wind

Soil erosions caused by water

- Sheet erosion
- Rill erosion
- Gully erosion
- Slip erosion
- Stream bank erosion

Soil erosions caused by wind

depends on the soil particle size

- Surface creep (5-10 mm)
- Saltation (1-1.5 mm)
- Suspension (less 1 mm)

DESERTIFICATION

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- **Desertification** refers to the persistent degradation of dryland ecosystems by climatic variations and human activities. The productive potential of the arid and semi-arid lands falls by ten percent or more
- Moderate desertification – 10-15 % drop in productivity
- Severe desertification – 25 -50 % drop in productivity

Effects of desertification

- Formation of barren lands & deserts
- Soil erosion
- Loss of biodiversity
- Loss of productivity of land

Causes of desertification

Deforestation, Overgrazing, Mining, and Quarrying

SOIL CONSERVATION PRACTICES

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1. **Till farming:** The land is plowed and broken up and smoothened. Topsoil becomes susceptible to erosion Special tillers break up and loosen the sub-surface soil without turning over the topsoil
2. **Contour farming:** Contour farming on gentle slopes, crops are grown in rows across the slop
3. **Terracing:** slops are converted into a series of broad terraces
4. **Strip cropping:** a strip of crops are followed with a strip of soil-saving cover crops like grass and grass-legume mixture
5. **Alley cropping:** crops are planted between trees and shrubs
6. **Shelter belts (or) windbreaks:** Trees are cultivated at land boundaries so that wind is blocked

FOOD RESOURCES

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The house is not a home, unless it contains food and fire for both the body and the mind - Benjamin Franklin

Sources

- Cultivated plants and domesticated animals,
- Agricultural crops
- Lives stock – milk, meat, poultry
- Fishes – Seafoods

- Green revolution
- White revolution
- Blue Revolution

World's food problem

- For the last 50 years food production was increased three times
- The Food and Agri. organization **(FAO)** of the United Nations estimated that 840 million people are chronically hungry, and 800 million people are in developing countries.
- In India, 40 % of the people suffer from malnutrition - children and pregnant ladies
- India is the third largest producer of staple crops, but still, 300 million Indians are undernourished.
- India has only half as much land as the USA, it has to feed nearly three times its population

PROBLEMS RELATED TO FOOD RESOURCES

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1. Over-grazing

- livestock graze on a particular piece of grassland, which surpasses the carrying capacity (sustainable capacity). The grazing pressure is higher than it carrying capacity, and the sustainability of the grazing land fails
- **Impacts of overgrazing:** land degradation, soil erosion, desertification, loss of biodiversity
- **Example:** Banni grasslands in Kutch

1. Land degradation

- Overgrazing removes vegetal cover—soil compacted
- Roots cannot go deeper into the soil
- Soil loss infiltration capacity—no water percolation
- Water runoff on the surface – leads to loss of soil fertility
- Nutrient cycle and organic cycles decrease—no decomposition
- Leads to organically poor, dry, and compacted soil

2. Soil erosion

When grasses are removed, soil become susceptible to the action of water and wind.

3. Loss of useful species

- Even the rootstocks which carry the reserve food for regeneration are also destroyed
- Other secondary species are hardier & fewer nutrients will grow. Nutritious juicy fodder is replaced by thorny species (Prosopis Juliflora)

PROBLEMS RELATED TO FOOD RESOURCES

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2. Agricultural practices

- **Slash and burn cultivation or shifting cultivation** - still among tribal people
- **Traditional agriculture**: More near to the natural conditions and results in low yield
- Impacts
    1. **Deforestation**- slash and burn – loss of forest cover
    2. **Soil erosion-** clearing the forest cover- loss of topsoil
    3. **Depletion of nutrients** – organic matter is destroyed, so nutrient decreases

MODERN AGRICULTURAL PRACTICES

Advantages:

1. High-yield varieties,
2. Hybrid seeds,
3. Single crop variety
4. High technical equipment
5. Energy subsidies that keep prices for farmers below market levels
6. Food production increased by the green revolution

Impacts:

1. Impacts related to a high-yield variety
2. Fertilizer related problems:
3. Pesticides related problems
4. Water logging
5. Salinity problem

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1. **Impacts related to a high-yield variety**
   - Monoculture crop pattern
   - Single pathogen can cause total devastation for large area
   - Hybrid varieties may lose their original nutrients and tastes
   - Gene modification
2. **Fertilizer-related problems**
   - Micronutrient imbalance
   - Eutrophication
3. **Nitrate pollution:** Blue baby syndrome or methemoglobinemia
4. **Pesticides related problems:**
   - **Pesticides** – Sulphur, Arsenic, Lead, and Mercury, Second generation pesticides (1939): DDT (Dichlorodiphenyltrichloroethane), Third: modified insects
   - Biomagnification, Bioaccumulation, Biotransformation
5. **Salinity:** Saline soil pH: 8.0, irrigation water has high salt content
6. **Water logging:** Pore spaces in the soil get fully drenched with water, and the soil-air gets depleted. The water table rise, and the roots lose their soil grip

EUTROPHICATION

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- natural or artificial addition of nutrients to the water bodies
- Causes
  - Natural runoff from topsoil
  - Weathering of Rocks
  - Run-off of inorganic fertilizer
  - Discharge of detergent and partially treated or untreated sewage
- Effects
  - Algal Bloom:
  - Harmful algal bloom: biotoxins released from phytoplankton
  - Red Tides
  - Hypoxia: reduced dissolved oxygen

TRANSFER OF BIOTOXINS IN TROPHIC LEVEL

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Bioaccumulation

The accumulation of contaminants by a species in concentration several orders higher than their surrounding.

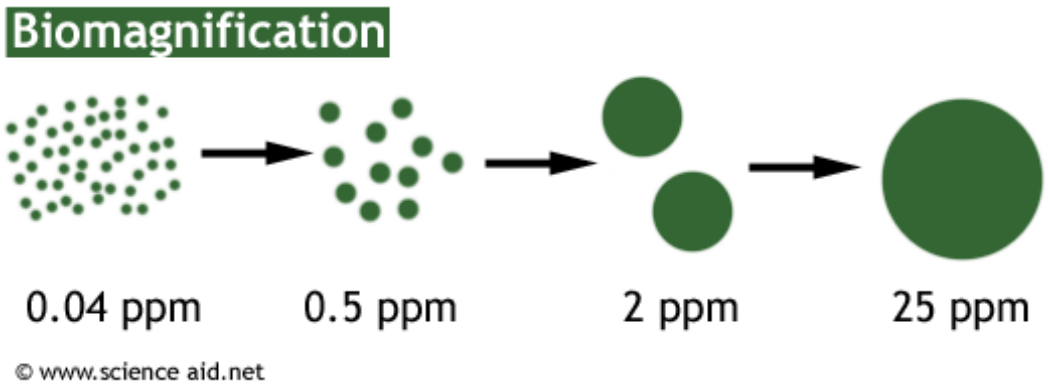
Accumulation of DDT and mercury

Bioconcentration

Is direct uptake by living organisms, and the concentration exceeds the ambient environmental concentration.

Biomagnification.

Tendency of the pollutant to concentrate as they move from one tropic level to next. E.g. Endosulfan can cause cancerous tumors, birth defects, and other developmental disorders.



FOOD RESOURCE MANAGEMENT

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1. Promote Sustainable Agriculture and Water Management
2. Crop Diversification
3. Technology Adoption
4. Training and Education
5. Post-Harvest Management
6. Market Access
7. Government Policies and Support
8. Climate Resilience
9. Rural Infrastructure
10. Community Engagement
11. Data Collection and Analysis
12. Awareness and Education

MINERAL RESOURCES

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- A **mineral** is a naturally occurring solid chemical substance that is formed through geological processes, and that has a characteristic chemical composition, a highly ordered atomic structure, and specific physical properties.
- **Ore**: a mineral or a combination of minerals
- Formed over a period of millions of years
- Non-renewable
- localized occurrence and in a limited quantity
- The quality also varies from place to place
- The presence fossil fuel is bigger compared to minerals

- **USES OF MINERALS**
- Development of industrial plants and machinery
- Technology process required metals and non-metals
- All industrial machines
- Constructions, developmental projects like road, and bridge constructions
- Defense equipment
- Communication
- Energy resources - fuels – uranium, coal, and lignite
- Jewellery

MINERAL RESOURCES

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Minerals are classified into

- **Metallic**—bauxite, hematite and
- **Non-metallic** – graphite, diamond
- **Critical minerals**–essential for the economy
- **Strategic minerals**– essential for defense

Major minerals of India

- **Energy-generating** minerals -- coal- lignite, and uranium
- **Commercially used** minerals – Al, Fe, and Cu

Mineral corporation of the Indian government

1. Mineral Exploration corporation ltd (MECL)
2. National Aluminium Company Ltd (NALCO)
3. Femnor mineral India
4. Bharath aluminum company Ltd
5. Hindustan zinc Company Ltd
6. Hindustan copper Company Ltd
7. Sikkim mining corporation

TYPES OF MINING

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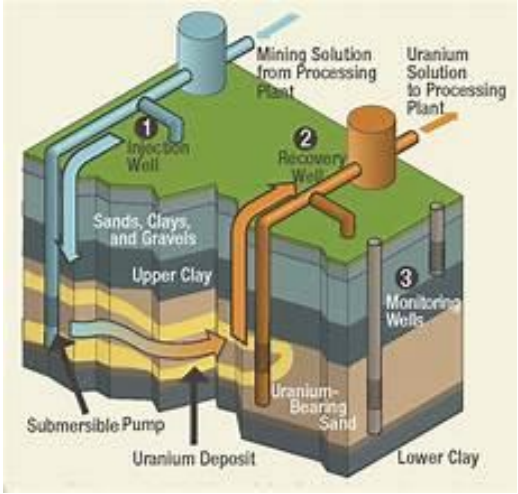
SURFACE MINING

best suited to extract minerals that are close to the surface of the earth. Cost-effective mining compared to underground mining. Common minerals extracted include coal, iron, and bauxite.

UNDERGROUND MINING

extract minerals that are located deep under the surface of the earth. Common minerals extracted include gold, lead, and silver. Depth can be as shallow as 300 meters to 3 km.



PLACER MINING

Placer mining is the method of separating valuable metals from sediments through sifting.
Often takes place in riverbeds, sands

IN-SITU MINING

recovering minerals by injecting a solution that dissolves the mineral and pumping back the solution Ex: uranium.

EFFECTS OF EXTRACTION AND USE OF MINERAL RESOURCES

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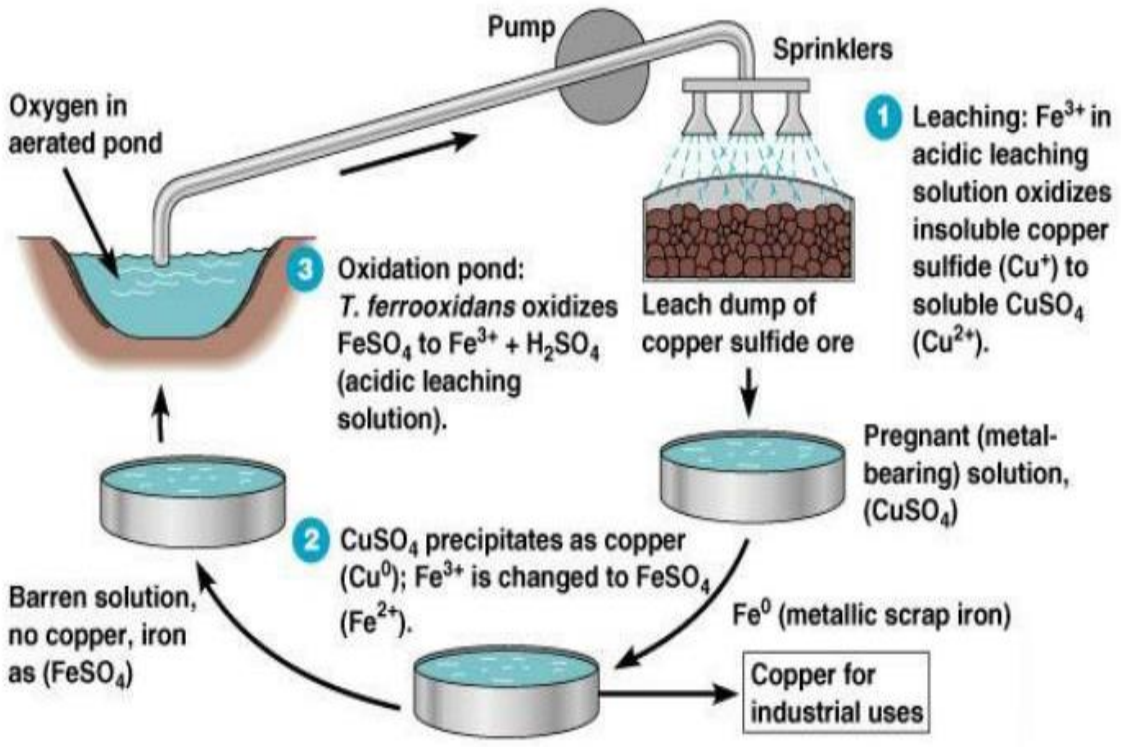
1. De-vegetation
2. Defacing of landscape
3. Subsidence of land
4. Groundwater contamination
5. Surface water contamination
6. Air pollution
7. Occupational health hazards
8. Mine accidence

Major mines known for causing severe problems

- **Jaduguda uranium mining** – radioactive hazard. Some people living near India's uranium mine in Jaduguda are suffering deformities.
- **Jharia coal mines** – Jharkhand – under mining –land subsidence and displacement of people
- **Sukinda chromite mines** –Orissa- heavy metal poisoning
- **Kudremukh iron ore mine** - Karnataka – river pollution
- **East coast –bauxite mine** – Orissa – land encroachment
- **North –East coalfield** – Assam – S contamination

REMEDIAL MEASURES

1. Adopting diff. techniques like
   Microbial leaching techniques – bacteria are used to consume a particular type of chemical
2. Mines can be re-vegetated by plants
3. Preventing toxic drainage discharge
4. Controlling air pollution
5. Resource Assessment and Planning
6. Mineral Conservation Policies
7. Reduce Waste and Losses
8. Land Rehabilitation and Reclamation
9. Water Pollution and human health assessment
10. Community Engagement and Benefit Sharing
11. Best Practices and Certification
12. Capacity Building
13. Monitoring and Reporting



CASE STUDY: LAMBAPUR-PEDDAGATTU PROJECT (NALGONDA, ANDHRA PRADESH)

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- Uranium Corporation of India Limited (UCIL) -Uranium mining in Lambapur and Peddagattu villages.
- To extract the ore pitchblende of 11 million tons of uranium in 20 years. 1 km from human habitats, 10 km from Nagarjuna Sagar Dam, and 4 km from Akkampalli reservoir, which is the drinking water source for Hyderabad people
- 20-year mining leaves 7.5 million metric tonnes of radioactive waste
- 445 ha of Yellapurum Reserve forest and the Rajiv Gandhi Tiger Sanctuary.
- **Saraswati Kavula** is an **Anti-nuclear activist** from Movement against Uranium Projects (MAUP), Andhra Pradesh
- Quality of ore in Andhra Pradesh is very poor: One ton of usable uranium from 3000 tons of ore is processed every day.
- Two crores and seventy lacks (27 million) tons of Radioactive waste in just one mine
- Contamination of water source (irrigation and drinking) for six districts.
- UCIL and other Nuclear establishments working under the DAE are not following any environmental and people safety precautions.
- They have no disaster preparedness plan, which should be in the public domain.

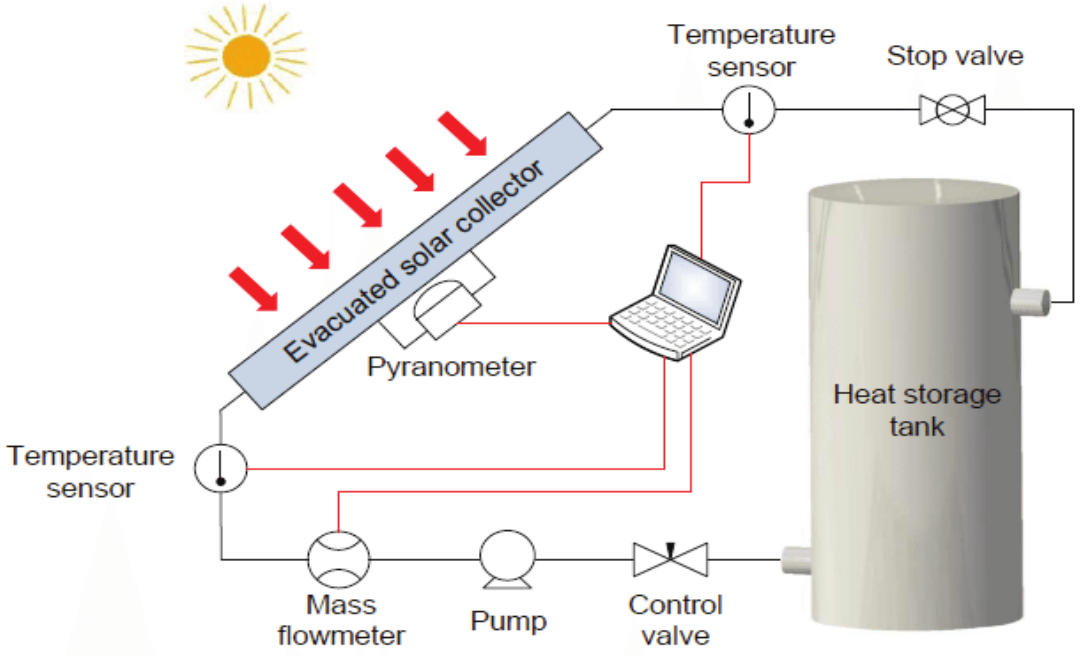
URANIUM CORPORATION OF INDIA LIMITED
UCIL
TUMMALAPALI.
PROHIBITED AREA.
UNDER SECTION 30,&27 OF ATOMIC ENERGY ACT - 1962
NO ENTRY WITHOUT PERMISSION

SOLAR ENERGY

Two ways two utilize

1. Conversion of solar energy into thermal: Solar collectors
2. Photovoltaic cell

Solar collector



Photovoltaic cell

- **Photo voltaic cells or Solar cells** are made of thin wafers of semiconductor materials like silicon and germanium.
- When solar radiations fall on them, a potential difference is produced, which causes the flow of electrons.
- The direct current gets converted to alternating current using inverters

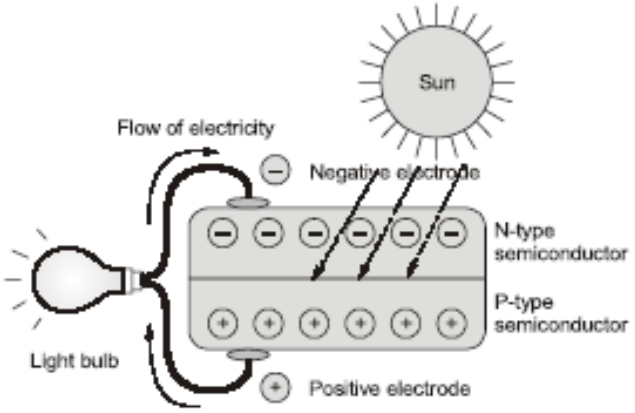
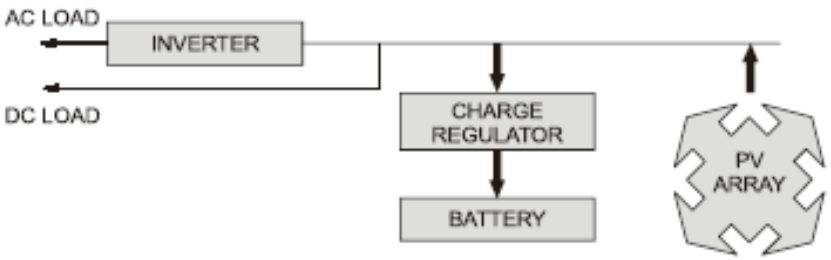
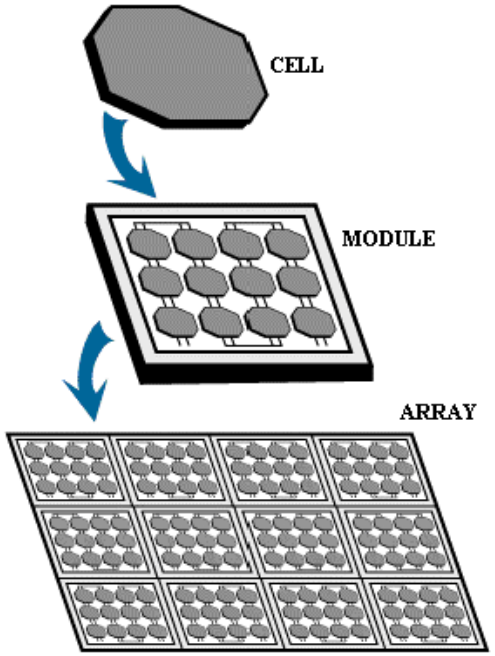
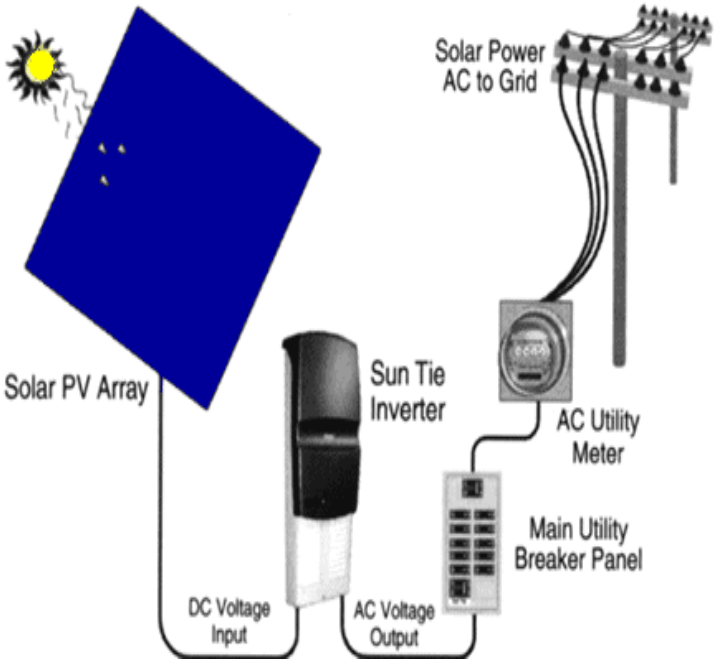


Fig. 1.2 (a): Working of a PV Cell



Typical output of a module (~30 cells) is ≈ 15 V, with 1.5 A current





SOLAR ARRAY IN INDIA

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Most parts of India have 300 - 330 sunny days in a year, which is equivalent to over 5000 trillion kWh per year - more than India's total energy consumption per year.

1. Bhadla Solar Park – 2,250MW. The country's biggest solar power plant is found in the state of Rajasthan
2. Shakti Sthala solar power project – 2,050MW.
3. Ultra Mega Solar Park – 1,000MW. ...
4. Rewa Solar Power Project – 750MW. ...
5. Kamuthi solar power plant – 648MW.

Year	Annual Solar power generation (TWh)
2013–14	3.35
2014–15	4.60
2015–16	7.45
2016–17	12.09
2017–18	25.87
2018-2019	39.27
2019–2020	**50.13**

APPLICATIONS OF SOLAR ENERGY

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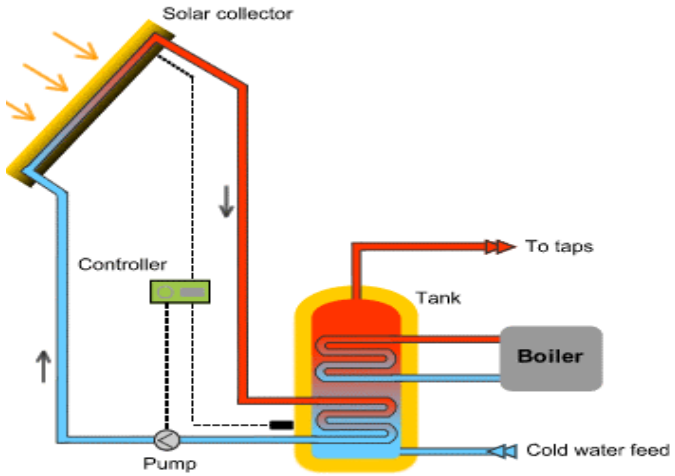
- Photovoltaic cells
- Solar thermal power generation
- Solar drying system
- Solar cooler
- Solar water heater
- Cooling and refrigeration

Solar cookers

Solar water heating systems



ADVANTAGES AND DISADVANTAGES OF SOLAR ENERGY

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Advantages

- It is clean, noise-free renewable energy resources
- Eco-friendly
- A typical solar cell can produce about 0.7 W of electricity
- Low maintenance cost as no moving parts
- Can be used in remote, inaccessible areas where power transmission is difficult and quite expensive

Limitations

- Solar cells can produce electricity only on sunny days
- Solar power station do not match the power output of the conventional stations
- Mostly used for charging batteries

WIND ENERGY

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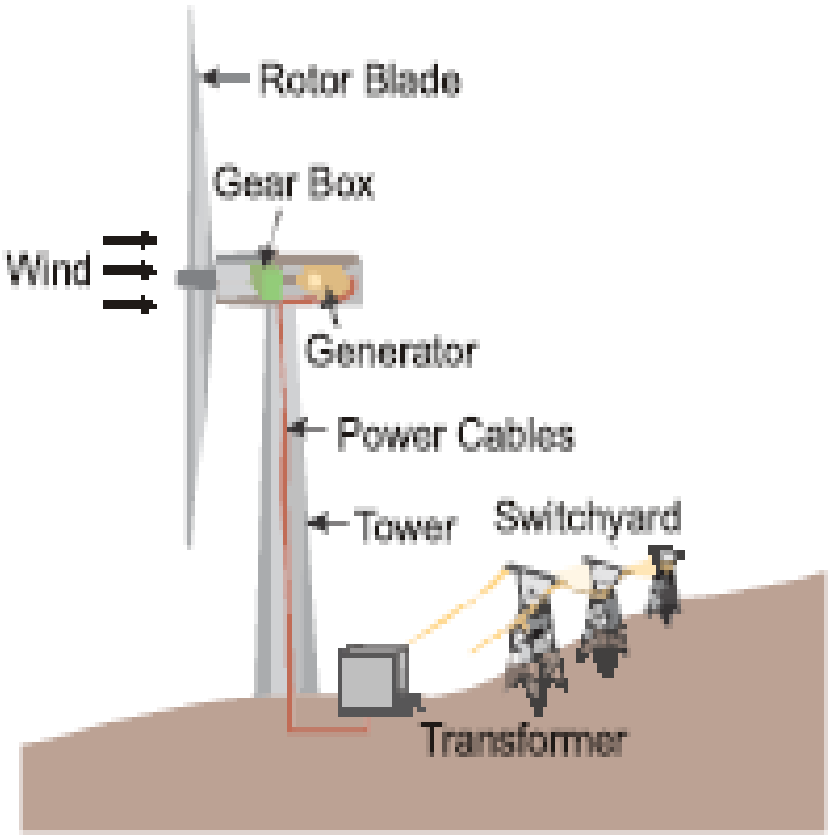
- kinetic energy generated by virtue of the movement of the large air mass caused due to the differential heating of the atmosphere.
- **Can be easily used in the grid, pumping, desalination, and telecommunication**
- **Influences more by terrain, water bodies, vegetation, and geographical region**
- **Wind Energy conversion system:** converts mechanical energy into electrical energy
- **Major constituent of WECS:** wind turbine, generator, interconnected apparatus and control system
- **WECS can operate in parallel with public and local grid**
- **APPLICATIONS:** To lift water from the ground or flooded mines, To grind cereals and grains, Generation of electricity



ADVANTAGES AND DISADVANTAGES OF WIND ENERGY

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Advantages

- Wind energy is fueled by the wind, so it's a clean fuel source.
- Wind energy is one of the lowest-priced renewable energy technologies available today. No recurring expenses
- Wind turbines can be built on farms, where most of the best wind sites are found. Farmers can continue to work the land because the wind turbines use only a fraction of the land. Wind power plant owners make rent payments to the farmer for the use of the land.

Disadvantages

Wind flowing with sufficient speed is required

- Speed and direction of the wind is seasonal
- Whole through the year, cannot expect the required output of electricity
- Installation cost is high. The technology requires a higher initial investment than fossil-fueled generators.
- Establishment of a wind farm requires a large land area of about 12 hectare for 1 MW
- Low energy density

WIND ENERGY IN INDIA

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- India’s potential is conservatively estimated **at 45,195 MW. India** is placed at the third position in the world in terms of new construction, and this corresponds to an overall increase of over 40% in new wind power stations
- **Wind power accounts for 6% of India's total** installed power capacity, and it generates 1.6% of the country's power
- India’s largest private power company, TATA Power, is setting up a 100 MW wind energy project in India. National Thermal Power Corporation plans to install a wind energy capacity of 250 MW. Hindustan Petrochemicals Company Limited and Oil and Natural Gas Commission are foraying into the wind energy segment with wind farms of 100 - 150 MW at various sites along India’s coastline.
- **Largest wind form in India, is located near Kanyakumari (Tamil Nadu, which generates 380 MW of electricity)**

Aralvaimozhi, the Muppandal wind farm which the largest in Asia is located near the village of Muppandal

Structure of Windmill, Germany

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GEOTHERMAL ENERGY

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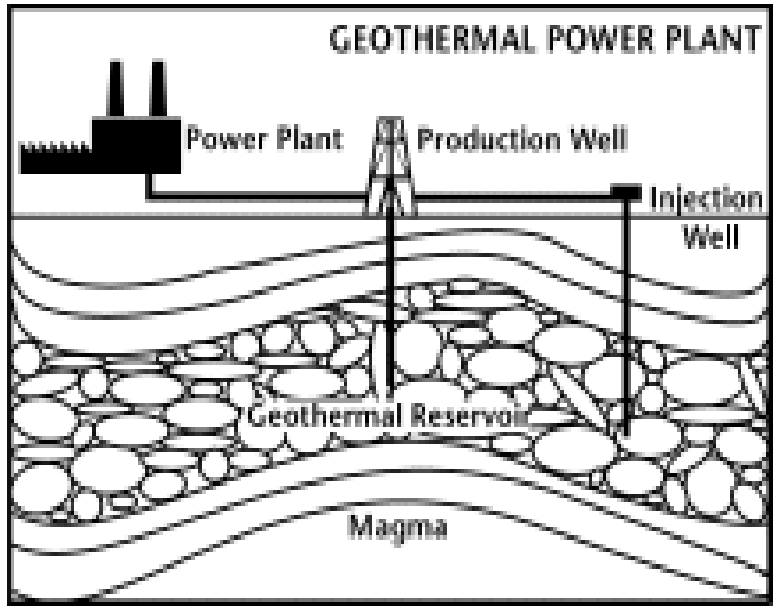
- Geothermal energy is a renewable thermal/ heat energy found in the **rock formation of the earth**. Possible manifestation: hot water spring, geysers
- When the mantle becomes melted, magma is created. The magma reaches the crust and heats nearby rocks and water. The heated water can reach the surface and form hot springs and geysers.
- Geothermal energy is an **underused heat and power resource that is clean (emits little or no greenhouse gases), reliable (average system availability of 95%), and homegrown (making us less dependent on foreign oil).**



GEOTHERMAL RESOURCES

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- **FUMAROLES**
- **RING OF FIRE**

ADVANTAGES AND DISADVANTAGES OF GEOTHERMAL ENERGY

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Advantages

- Low running cost
- It is a clean fuel source with no pollution.
- Easy and modular construction

Disadvantages

- Poor efficiency compared to fossil fuel plants (15 %)
- Large amount of withdrawal can cause surface subsidence
- Hot water can contain gases like radon, ammonia, and hydrogen sulphide.
- Large area is needed
- Available only in some selected areas.

OCEAN ENERGY

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- Indirect form of solar energy
- **Sources:** Tide, Waves, and Ocean Thermal Energy Conversion (OTEC)
- **Wave Energy**
  - **Affected by/ Factor governing**:
    - Wind speed
    - Fetch value (the uninterrupted distance on the ocean over which wind can below before reaching the point of reference): 5-45 km
    - Depth of seawater
  - Wave energy is more efficient than wind energy, as waves are formed by the concentration of wind energy
  - *Empirically, wave energy is defined as the rate at which water is transferred across one-meter line at the right angle to direction.*
  - $P=0.96H^2T$

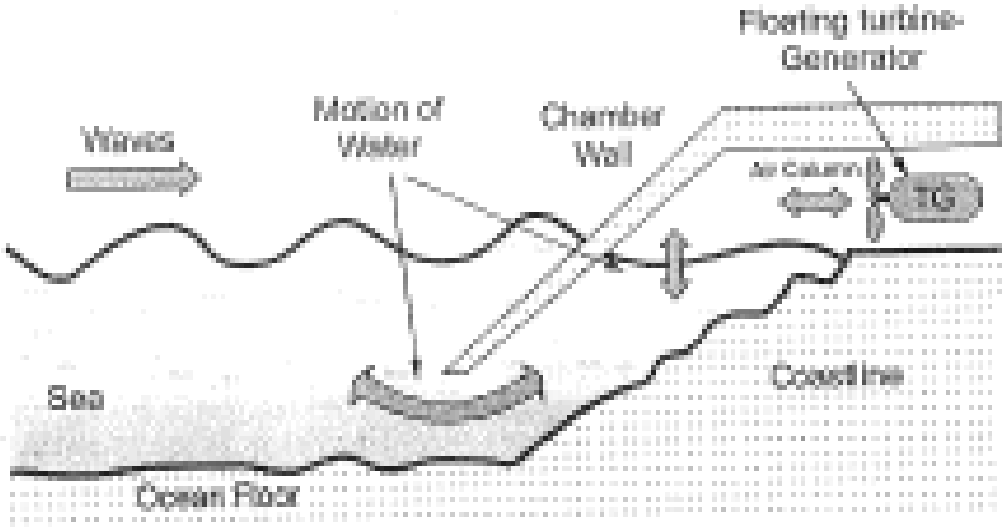

Fig: 1.14 Wave Energy Conversion System

ADVANTAGES AND DISADVANTAGES OF WAVE ENERGY

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Advantages

- More energy efficient than wind energy
- Do not require large landmass

Disadvantages

- Capital and operational cost is high
- Limited zone of capture.
- Potential to impact marine biodiversity
- Highly variable.

TIDAL ENERGY

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- Energy is extracted from tides, which are generated due to the gravitational effect of the sun and moon on the Earth.
- **The height of the tide above and below the tidal basin result in the movement of the turbine**
- Need a suitable **bay area** to store water and release it during low tide. The bay area is defined by **tidal range.**

Components

1. Barrage/ Dam / Dykes
2. Sluices:
3. PowerHouse:

Advantage

1. No dependent on seasonal variation, predictable
2. Low operational cost

Disadvantages

1. Capital is high
2. Silting of the basin is common.
3. Efficiency depends on turbine and tidal range.

OCEAN THERMAL ENERGY CONVERSION

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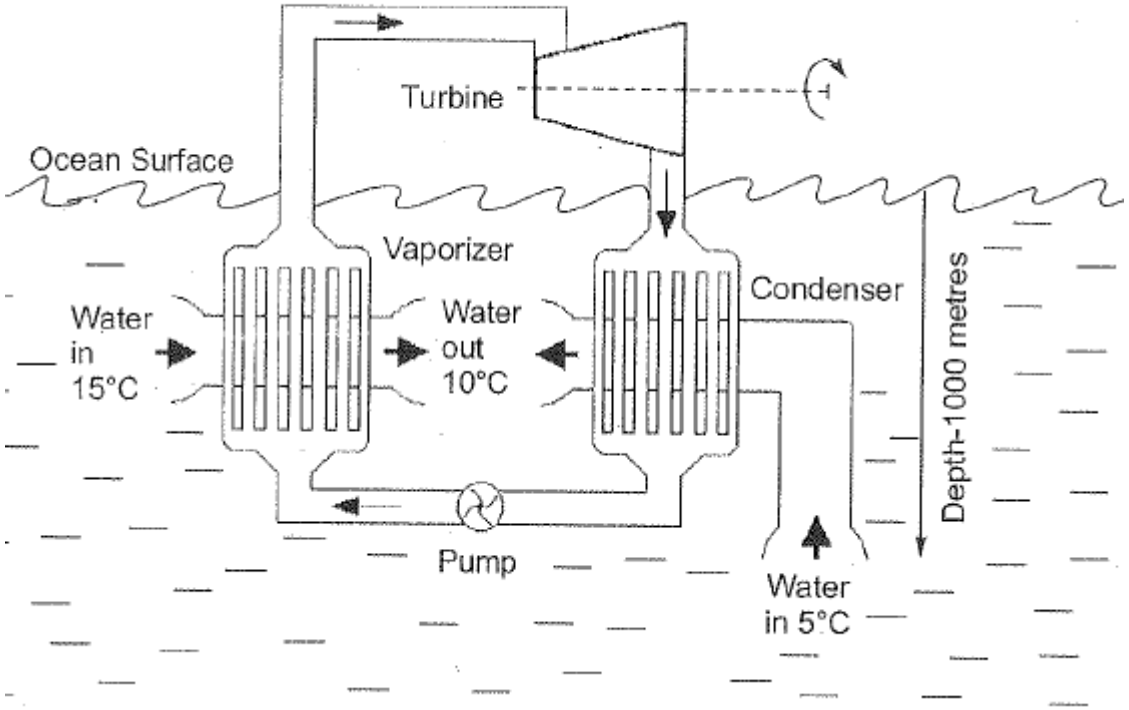
- Makes use of the temperature difference between the surface water of the ocean and the depth of the ocean
- Harnessed in tropical oceans (28 to 5 degrees C)
- A difference of 20° C and more is required to run the OTEC power plants
- Principle:
- The warm surface is used to vaporize the liquids like ammonia to run a heat engine, and the deep water is used to condense the vapors
- The energy carrier, seawater, is free, although it has an access cost associated with the pumping materials and pumps energy costs



ELECTRICITY PRODUCTION

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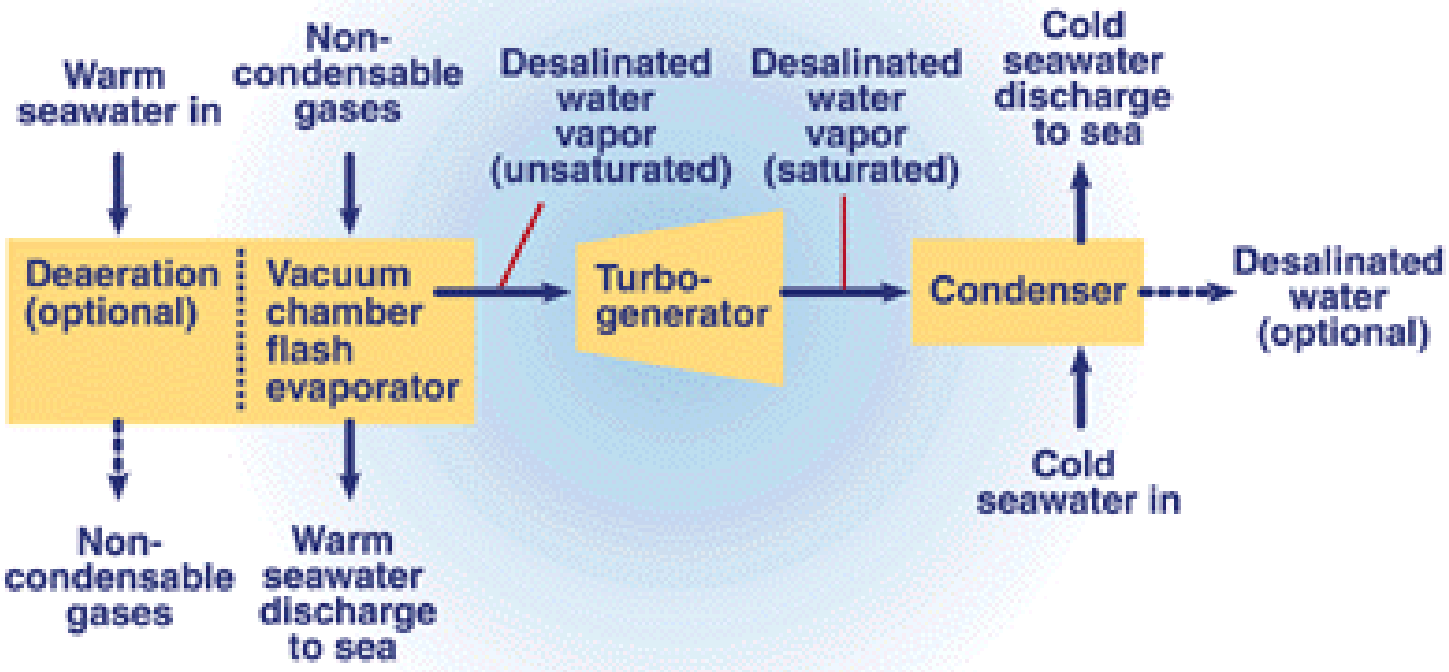
2. Open cycle



- Water is the working fluid
- Desalinated water can be produced

FACTORS TO BE CONSIDERED WHILE CHOOSING A OTEC SITE

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- Thermal gradient in the ocean
- Topography of the ocean floor
- Meteorological conditions – hurricanes
- Seismic activity
- Availability of personnel to operate the plant
- Infrastructure – airports, harbors, etc.
- Local electricity and desalinated water demand.
- Political, and ecological constraints
- Cost and availability of shoreline sites
- India OTEC potential is 180, 000MW due to Exclusive Economic Zone and Continental shelf.

ADVANTAGES AND DISADVANTAGES OF OTEC

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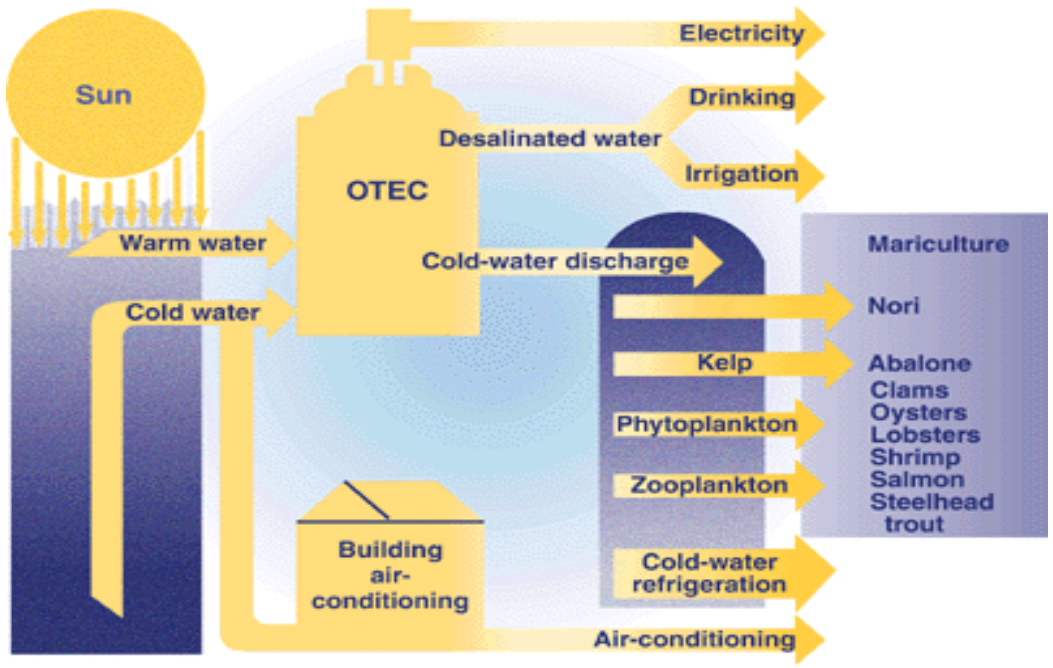
Advantages

- Produces desalinated water
- Provides air-conditioning for buildings
- Provides moderate-temperature refrigeration
- Especially beneficial for small islands
- Promotes mariculture
- Floating OTEC can provide power to offshore mining and processing

Limitations:

- Low-grade solar energy with poor energy recovery efficiency
- Floating plants – navigational hazard
- Chlorine used for preventing biofouling – hazardous
- Metal pieces entrained – affect marine organisms.
- Mixing of warm and cold seawater affects dissolved oxygen



BIOMASS ENERGY

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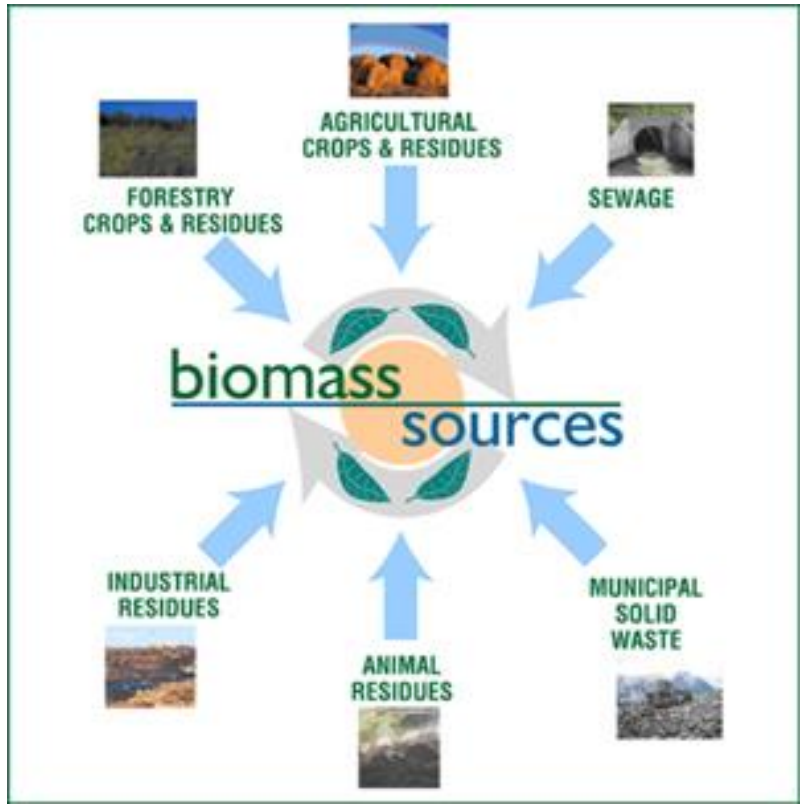
- Biomass refers to **solid carbonaceous material** developed from plants and animals. Examples of biomass include wood, leaves, animal waste, crops, and bones.
- Biomass energy is the **utilization of energy stored in organic matter.**
- Considered **carbon-neutral energy source**
- Biomass is stored solar energy that can be converted to electricity and fuels such as methane, ethanol, and bio-diesel.



- Biomass energy is extracted as:
  - **By direct burning of dry plants**
  - **By fermentation**: Ethanol, exclusively from corn.
  - **Petro crops**: Euphorbias and oil palms are rich in hydrocarbons
  - **Agri. & urban waste biomass**: Crop residue, bagasse, coconut shells, peanut hulls, cotton stalks, animal waste, fishery, and poultry waste used as **charcoal, briquetting (3500 kcal/ kg)**

BIOMASS POWER CONVERSION

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- Convert biomass into heat and electricity
- Process similar to fossil fuel
- Biomass is burnt to heat a boiler. The steam is directed towards the turbine. The charged magnetic field produced a current
- Other conversion techniques
  1. **Densification:** bulky biomass is reduced to volume-to-weight ratio by compression at controlled temperature and pressure
  2. **Combustion**
  3. **Incineration:** Burning at high temperatures, mostly waste
  4. **Thermochemical conversion:** Pyrolysis, Gasification and Liquefication
  5. **Biochemical Conversion**: Anaerobic digestion and Ethanol fermentation

THERMOCHEMICAL CONVERSION

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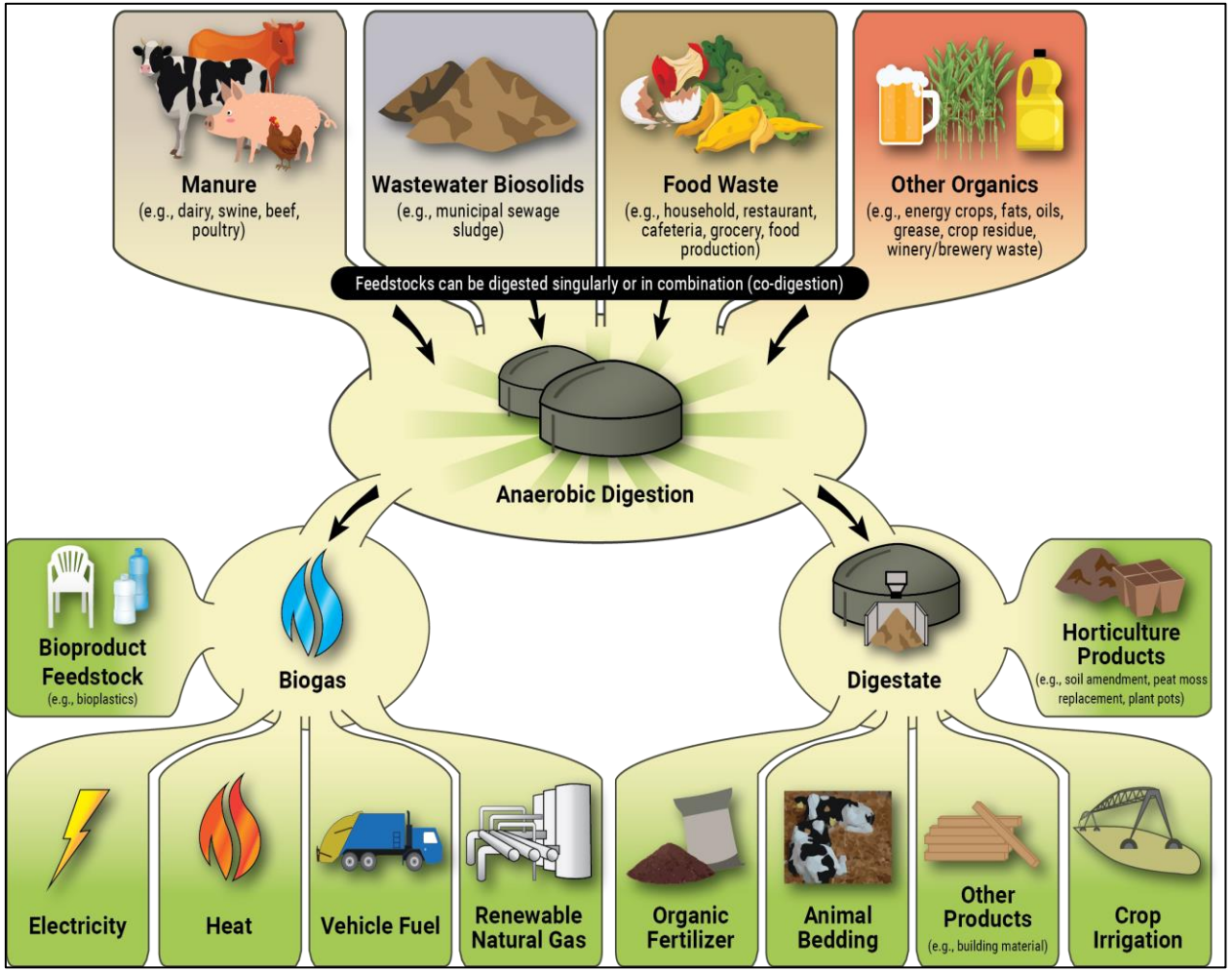
PYROLYSIS

- **Thermal decomposition without oxidizing agents like CO2, oxygen, or steam.**
- Temperature range 300 and 850°C.
- **Solid:** a **char-like substance** (10–35 MJ/kg) and 20–50%,
- **: Ash (10–50%).**
- **Liquid:** a complex mixture of hydrocarbons (5–15 MJ/kg) and 30–50%.
- **Gas:** a mixture **of CO, CH4, CO2, H2, and other volatile waste constituents**. The heating value and gas yield may be around 3–12 MJ/Nm3 and 20–50%, respectively.

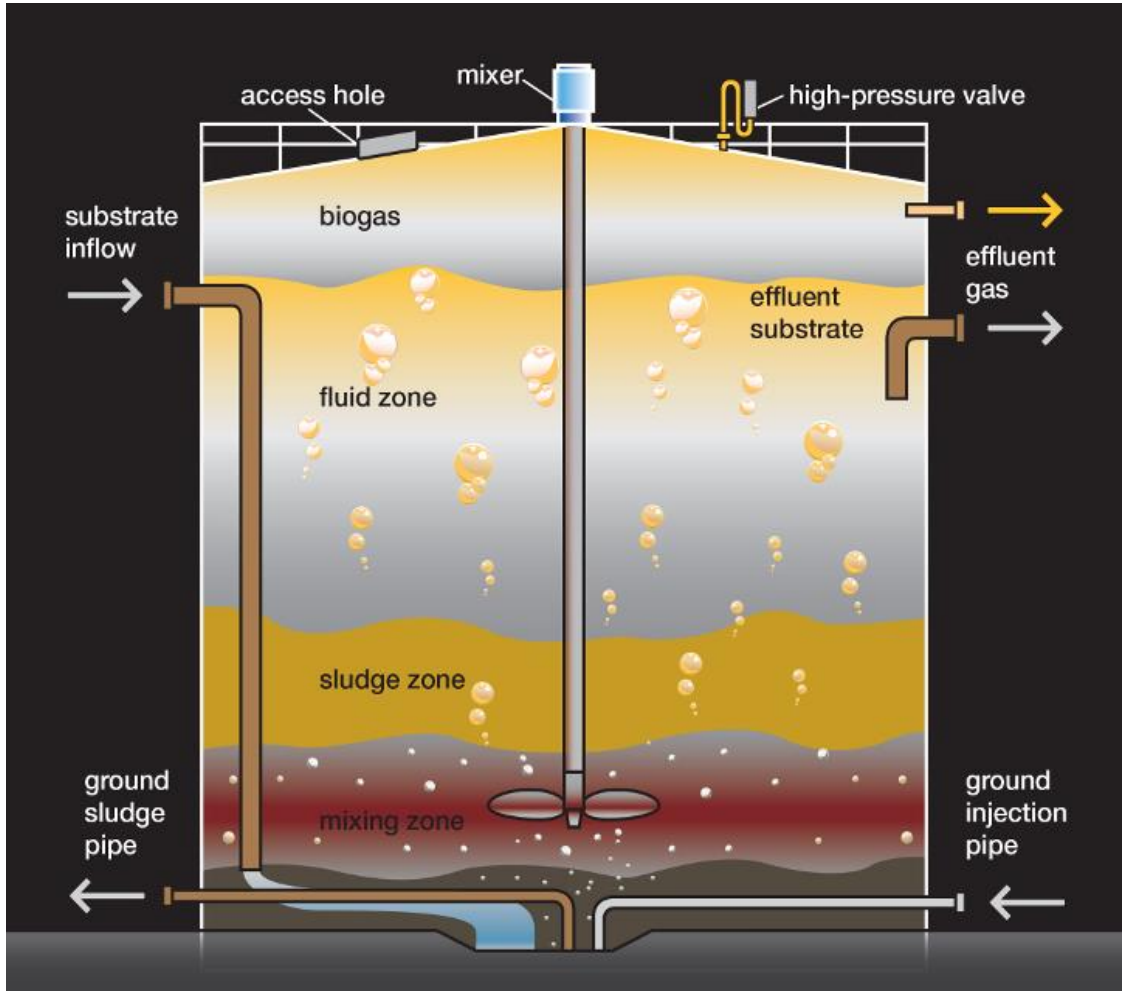
GASIFICATION

- Chemical and thermal conversion of carbon-based materials into a primarily gaseous output air, oxygen, or steam.
- **Temperature range from 800 to 1100°C when air as an oxidant,**
- **1500°C when using oxygen**.
- **Solid:** Ashes 30–50% of the input weight.
- **Liquid:** Oil or tar, 10–20%
- **Gas:** Higher CO_2 fractions (3–12 MJ/Nm$^{3)}$, 30 to 60%

Anaerobic Digestor



Inflow and outflow for Anaerobic Digestion Source: US EPA



Schematic of Anaerobic Digestor Source: Internet

BIOCHEMICAL CONVERSION

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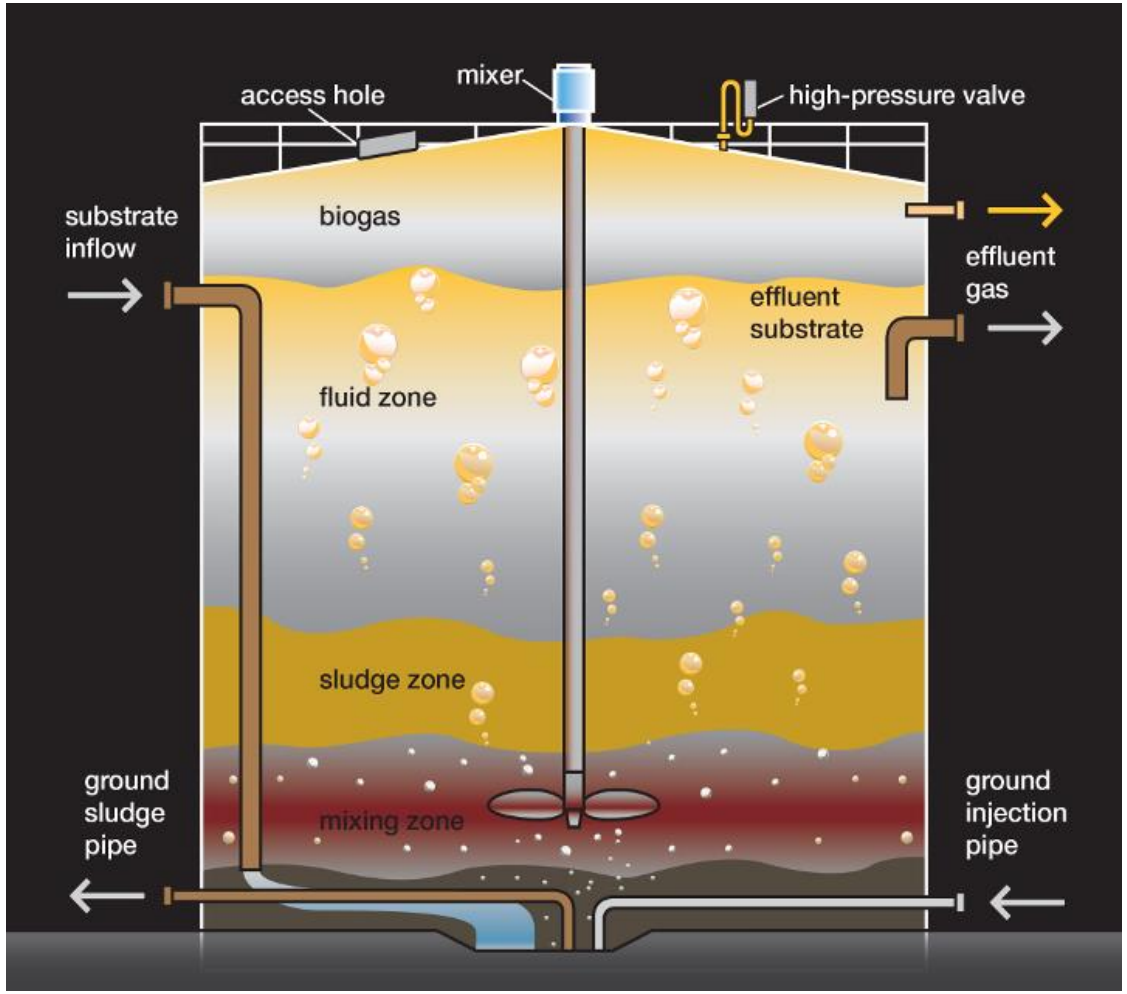
Anaerobic Digestion

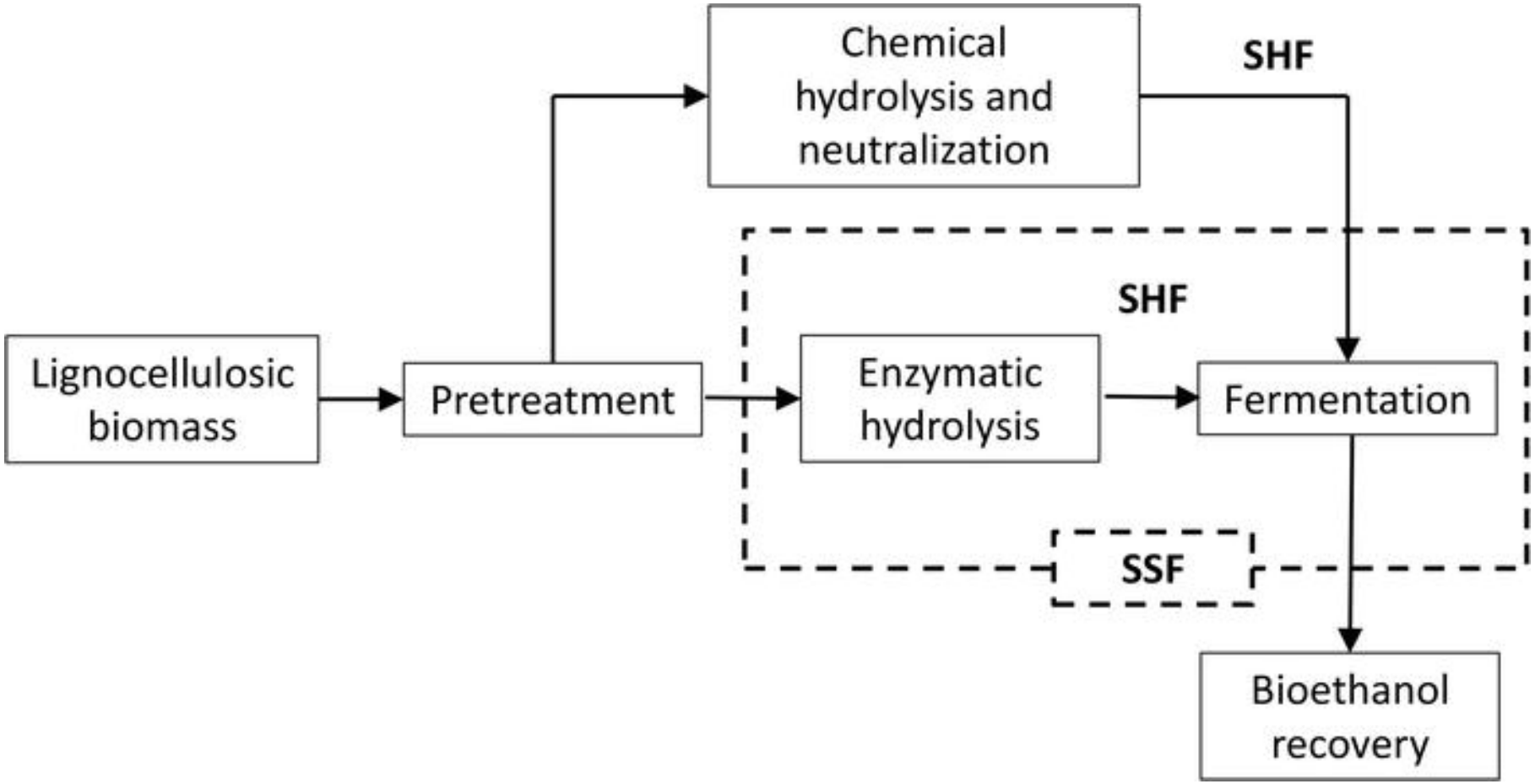
- Process through which bacteria break down organic matter in the absence of free oxygen.
- Primary objective – Energy Production.
- Biomethanation is anaerobic digestion of organic materials which is converted into biogas.
- 3-stage process: large organic polymers are first hydrolyzed and then fermented into short-chain volatile fatty acids (VFAs).
- These acids are then converted into methane and carbon dioxide.
- Methane - fuel (biogas) and value added products (bioplastics).
- Major drawback- Sensitive process and need of proper maintenance.



Schematic of Anaerobic Digestor Source: Internet

Ethanol Fermentation

- Decomposition of biomass containing sugar like sugarcane, potato, etc



BIOGAS

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- Produced by biological digestion of waste under anaerobic conditions at a temperature range of 35-70 degrees C.
- Biogas comprises primarily methane (CH_4) and carbon dioxide (CO_2) and may have small amounts of hydrogen sulfide (H_2S), moisture, and siloxanes.
- Calorific value 5000 to 5500 kCal/ kg
- Produced during anaerobic digestion
- Factor affecting
  - Solid-to-water ratio
  - Temperature
  - Seeding
  - pH
  - Carbon to nitrogen ratio (30:1)
  - Stirring

GOBAR GAS PLANT

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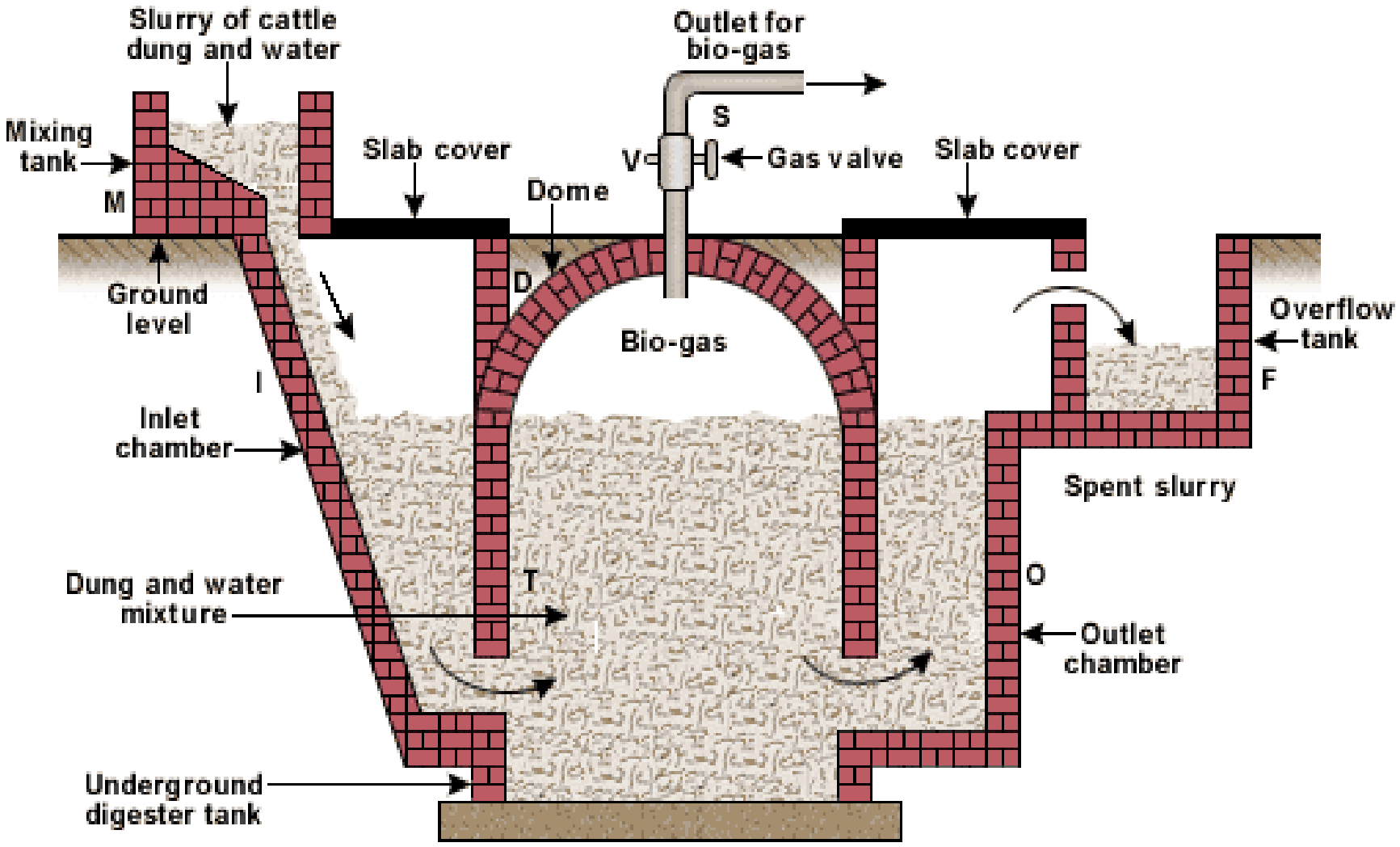

Fixed-dome type bio-gas plant.

BIO-FUELS

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- Liquid fuel recovered from agricultural and forest crops and residue

- TYPES OF BIOFUELS
  - **Primary:** used in unprocessed form such as fuel wood, chips, etc.
  - **Secondary:** Material resulting from processing of Biomass

- GENERATIONS
  - **First:** conventional biofuels made from sugar or starch. Food crop is used directly
  - **Second:** produced from non-food crops such as waste, woods
  - **Third:** made from engineered crops such as algae
  - **Fourth:** produced from hydroprocessing
- Biodiesel
- Produced from non-edible oil such as Jatrops, Karanja. Raw oil was subjected to transesterification.
- Biodegrabale, produce less CO_2, and less SO_2 emissions., Neat fuel, Higher flash point

HYDROENERGY

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- Energy of moving water obtained when a descent of the river is compressed to a single location.
- Small hydropower: produce electricity upto 25 MW
- Large hydropower
- Advantages
  - Low operating and maintenance cost
  - Short gestation period
- Disadvantages
  - Need areas with sufficient supply of water
  - Floods in low lying area
  - Emission of Green house gas

RENEWABLE ENERGY vs GREEN ENERGY

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RENEWABLE ENERGY	GREEN ENERGY
Relies on resources that restore over a short period of time	Subset of renewable energy that provide highest benefit and do not impact the environment
Can have high environmental impacts	

NON-RENEWABLE RESOURCES

- Coal
- Petroleum
- Natural Gas
- Nuclear Energy

CHEMCIAL SOURCES OF ENERGY

- Fuel Cell
- Hydrogen

EQUITABLE USE OF RESOURCES FOR SUSTAINABLE LIFESTYLES.

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- Sustainable living means understanding how our lifestyle choices impact the world around us and finding ways for everyone to live better and lighter.
- **Sustainable Development Goals (4 Education and 12.8 Responsible Consumption).**

- Most people do not wake up with the intention to harm the environment - nor to help it.
- People do not change behavior based on what they should do.
- Sustainability is not the defining criteria. Even the people who want to live more sustainably often lack information and access to affordable and desirable products and services.
- Governments and business (who are also consumers!) to provide more information and support positive behavior change.

- Research institutions and non-governmental organizations – to develop sustainable lifestyle frameworks and future forecasting.
- Educational institutions – to raise awareness and critical thinking to connect how daily living decisions affect the world and people around us.
- Governments – to provide guidelines, frameworks and future forecasts, to support the adoption of sustainable policies and improve infrastructure.

Role of an individual in the conservation of natural resources.